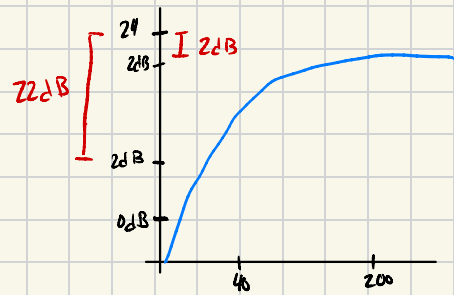


High - Pass Filter Design



$$\left| \frac{z^2 - 2}{\log_{10}(40) - \log_{10}(200)} \right| \leq 20_n$$

$$n = 2$$

$$\alpha = \frac{1}{Q}$$

$$H(s) = \frac{H_0 s^2}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2} \text{ for } K=1$$

(Chebyshev)

$$\rightarrow H(s) = \frac{s^2}{s^2 + 1.097734s + 1.102510}$$

$$\omega_0 = \sqrt{1.102510}$$

$$\left(\frac{\omega_0}{Q} \right)^{-1} = (1.097734)^{-1}$$

$$Q = 0.956520215$$

$$\frac{Q}{\omega_0} = \frac{1}{1.097734}$$

$$Q = \frac{\omega_0}{1.097734} = \frac{\sqrt{1.102510}}{1.097734} = 0.956520215$$

$$H_{HPF}(j\omega) = \frac{V_{out}}{V_{in}} = \frac{K s^2}{s^2 + \left(\frac{R_1 C_1 + R_2 C_2 + R_2 C_2 (1-K)}{R_1 R_2 C_1 C_2} \right) s + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$Q = \frac{\omega_n R_1 R_2 C_1 C_2}{R_1 C_1 + R_1 C_2 + R_2 C_2 (1-K)}$$

$$\omega_n = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$\rightarrow \omega_n = \frac{1}{\sqrt{R^2 C^2}} = \frac{1}{RC} \rightarrow 2\pi f_n = \frac{1}{RC}$$

$$f_n = \frac{1}{2\pi RC}$$

Bad Approach

$$R_1 = R_2 = R \quad C_1 = C_2 = C$$

$$Q = \frac{\omega_n R^2 C^2}{2RC} = \frac{\omega_n RC}{2}$$

Good Approach

$$R_2 = m R_1, \quad R_1 = R \quad [k=1]$$

$$C_2 = n C_1, \quad C_1 = C$$

$$Q = \frac{W_n R (m R) C (n C)}{R C + R (n C)} = \frac{W_n R^2 C^2 m n}{R C (1+n)} = \frac{W_n R C m n}{(1+n)}$$

$$W_n = \frac{1}{\sqrt{R (m R) C (n C)}} = \frac{1}{\sqrt{R^2 C^2 m n}} = \frac{1}{R C \sqrt{m n}}$$

With Changes $\rightarrow f_n = \frac{1}{2\pi R C \sqrt{m n}}$

find m and n

$$Q(W_n) = \frac{\left(\frac{1}{R C \sqrt{m n}}\right) R C m n}{(1+n)} = \frac{\left(\frac{m n}{\sqrt{m n}}\right)}{(1+n)} = \frac{\sqrt{m n}}{(1+n)}$$

$$\sqrt{m n} = Q(1+n) \rightarrow m = \frac{(Q(1+n))^2}{n}$$

$$Q = 0.956520215$$

$$\text{let } n = 10$$

$$m = \frac{(0.956520215)(1+10)^2}{10} = 11.07066415$$

$$C = 68 \text{ nF} = C_1 = 68 \text{ nF}$$

$$\begin{aligned} C_2 = n C &\rightarrow C_2 = 10(68 \text{ nF}) \\ &= 10(68 \cdot 10^{-9}) \\ &= 68 \times 10^{-7} \text{ F} \\ &= \underline{0.00000068} \rightarrow 680 \text{ nF} \end{aligned}$$

$$R = ?$$

$$R_2 = m R_1, \quad R_1 = R$$

$$f_n = \frac{1}{2\pi R C \sqrt{m n}}$$

$$200 = \frac{1}{2\pi (R) (68 \cdot 10^{-9}) \sqrt{(10)(11.07066415)}}$$

$$R_{200} = \frac{1}{R (0.000004495)} \cdot R$$

$$\frac{R_{200}}{200} = \frac{1}{(0.000004495)} \cdot \frac{1}{200}$$

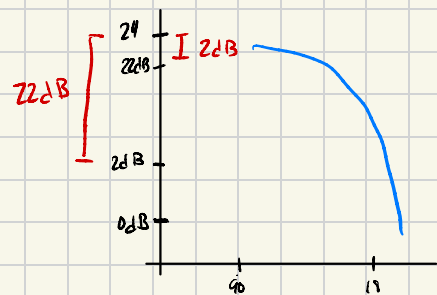
$$R = \frac{1}{200 (0.000004495)} = 1112.347052$$

$$R_1 = R = 1112.347052 \Omega \rightarrow 1.1 \text{ k}\Omega$$

$$R_2 = m R_1 \rightarrow (11.07066415)(1112.347052) = 12314.42063$$

$$\rightarrow 12.314 \text{ k}\Omega$$

Low Pass filter Design



$$\left| \frac{z^2 - z}{\log_{10}(40) - \log_{10}(18)} \right| \leq 20n$$

$$n = \frac{20}{20} = 1.43$$

$$n = 2$$

let $K=1$
Chebyshev

$$H(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$\rightarrow H(s) = \frac{1.102510}{s^2 + 1.097734s + 1.102510}$$

$$\omega_0 = \sqrt{1.102510}$$

$$\left(\frac{\omega_0}{Q} \right)^{-1} = (1.097734)^{-1}$$

$$Q = 0.956520215$$

$$\frac{Q}{\omega_0} = \frac{1}{1.097734}$$

$$Q = \frac{\omega_0}{1.097734} = \frac{\sqrt{1.102510}}{1.097734} = 0.956520215$$

